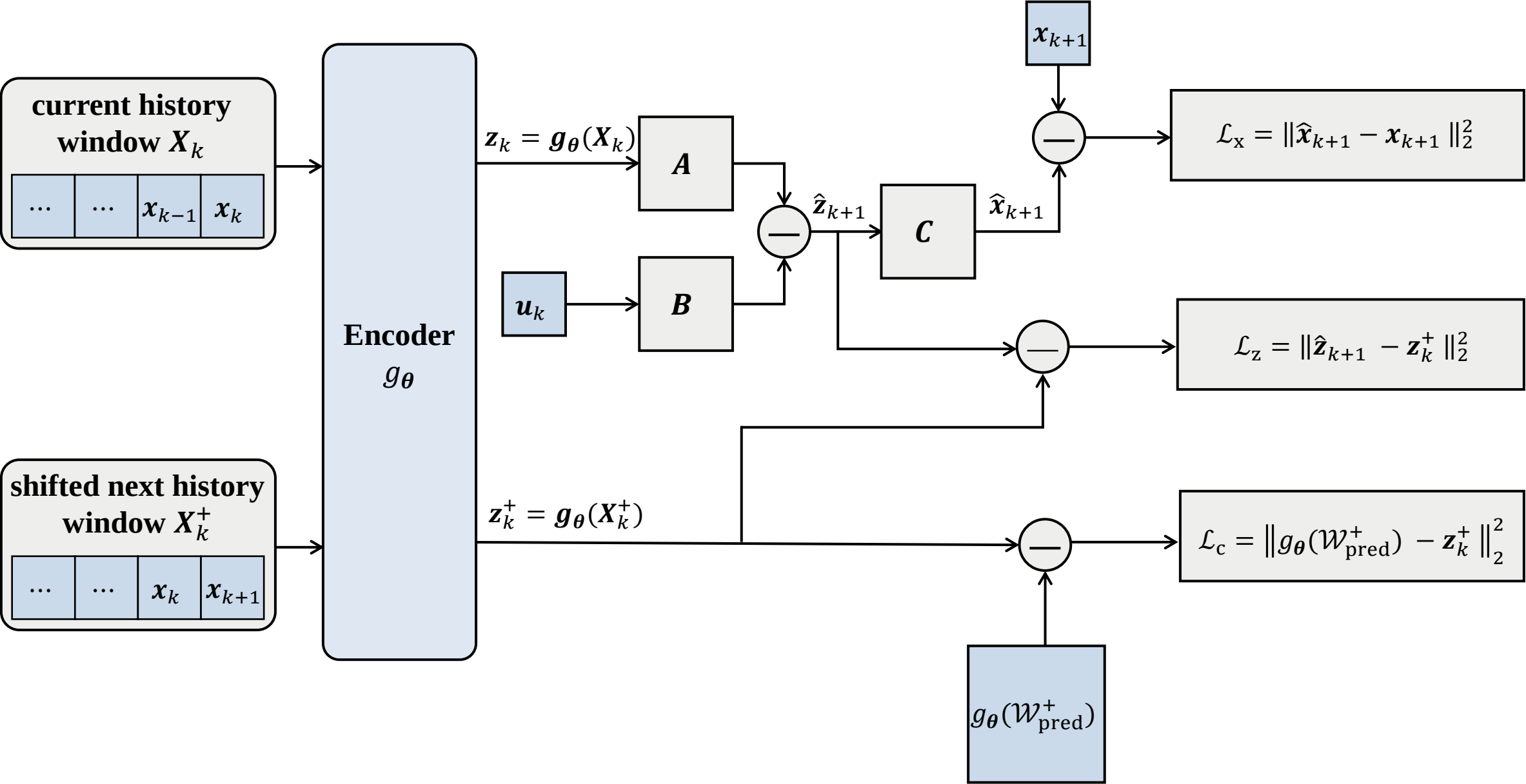

current history window X_k
… … x_{k-1} x_k
shifted next history window X_k^+
… … x_k x_{k+1}
Encoder g_θ
$z_k = g_\theta(X_k)$
$z_k^+ = g_\theta(X_k^+)$
u_k
A
B
C
$\hat{z}_{k+1}$
$\hat{x}_{k+1}$
x_{k+1}
$\mathcal{L}_x = \|\hat{x}_{k+1} - x_{k+1}\|_2^2$
$\mathcal{L}_z = \|\hat{z}_{k+1} - z_k^+\|_2^2$
$\mathcal{L}_c = \|g_\theta(\mathcal{W}_{pred}^+) - z_k^+\|_2^2$
$g_\theta(\mathcal{W}_{pred}^+)$